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BEDDING SYSTEMS AND METHODS

Abstract

Inventions of the present disclosure may be considered a bedding characterized by enhanced comfort, durability, long life, and decreased cost. The bedding may include sheeting, which may include pillowcases, pillow shams, a top sheet, a fitted sheet, blankets, and/or a duvet cover, and/or any combination thereof. The sheeting may include a unique weave pattern, which includes various techniques such as plain, dobby, Jacquard, percale, twill, oxford, drill, satin, and/or sateen. The fabric composition may be 100% polyester. A blend of a polyester and micro-polyester warp yarns may be included. Inventions may include a woven bedding fabric, a method for weaving a bedding fabric, and/or a system for a bedding fabric. The woven bedding fabric may be 100% polyester and have a different woven top pattern and bottom pattern. This combination of materials/weaving techniques results in bedding that is not only comfortable and aesthetically pleasing but also durable and cost-effective.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to the field of bedding, and more specifically to the field of bedding for enhanced comfort, durability, long life, wrinkle resistance, breathability, wicking away of moisture, and/or decreased cost and efficiency, and more specifically to bedding systems, methods, and/or kits.

BACKGROUND

[0002] Traditionally, bedding has been designed with a focus on basic comfort and warmth. Materials such as cotton, wool, and down have been commonly used in the manufacture of bedding due to their natural comfort properties. Over time, advances in materials science have led to the introduction of synthetic fibers and memory foam, which offer enhanced comfort, provide better support, and improved temperature regulation.

[0003] With the advent of synthetic materials, the bedding industry saw the introduction of polyester, polyester blends and other man-made fibers. These materials offer some benefits over more traditional approaches such as improved durability, ease of maintenance (e.g., machine washability), wrinkle resistance, breathable, moisture wicking and cost-effectiveness. However, such improvements often still fall short in terms of breathability and moisture-wicking properties compared to natural materials.

[0004] The introduction of memory foam and latex represented a significant advancement in bedding, particularly in mattresses and pillows. These materials provided enhanced support by conforming to the body's shape and reducing pressure points. Yet, issues such as heat retention in memory foam and the potential for latex allergies in some users have been noted. Recognizing the problem of heat retention in memory foam, manufacturers developed gel-infused foams and other materials designed to regulate temperature more effectively. Phase-change materials (PCMs) and breathable fabrics also emerged attempting to provide better thermal comfort while in use.

[0005] Advances in ergonomics led to bedding products tailored to different sleeping positions and personal preferences. This included adjustable mattresses and pillows, as well as designs catering to specific health conditions like chronic pain or spinal alignment issues. The integration of technology into bedding brought about smart mattresses and pillows capable of monitoring sleep patterns, adjusting firmness, and even interacting with other smart home devices. These innovations aimed to enhance the user's sleep quality through personalization and data-driven insights. There has been also a growing emphasis on sustainability in the bedding industry. This includes the use of organic materials, eco-friendly manufacturing processes, and products designed with a lower environmental impact in mind.

[0006] However, despite these advancements, existing bedding solutions often fail to address certain user needs. Common problems include inadequate support for different body types, poor temperature regulation leading to overheating or feeling too cold, and issues with durability and ease of maintenance. Furthermore, traditional bedding designs have not adequately adapted to the

diverse needs of individuals with specific health conditions, such as chronic pain or allergies.

[0007] Challenges remain to balance concerns with integrating advancements and new technologies with maintenance of traditional benefits and sustainability. Additionally, improvement is still desirable, and Applicant introduces significant advancement in the field of bedding related to enhanced comfort, personalization health benefits, durability and ease of use over traditional bedding solutions. This positions the inventions of the present disclosure as a valuable contribution to the bedding industry, addressing both longstanding and emerging needs of users. It is to these and additional challenges that the inventions of the present disclosure are directed.

[0008] Therefore, Applicants desire bedding, systems, methods, and kits to improve bedding comfort, durability, lifespan, cost efficiency, manufacturing ability and efficiency, without the drawbacks presented by the traditional systems and methods.

SUMMARY

[0009] A system, method, and/or kit for bedding is disclosed. In some examples, a bedding system, method, and/or kit may be characterized by enhanced comfort, durability, long life, and decreased cost is disclosed. This Summary is provided to introduce a selection of disclosed concepts in a simplified form that are further described below in the Detailed Description including the drawings provided. This Summary is not intended to identify key features or essential features of the claimed subject matter. Nor is this Summary intended to be used to limit the claimed subject matter's scope. Further, features or variations may be provided in addition to those set forth herein. For example, embodiments may be directed to various feature combinations and sub-combinations described in the summary and detailed description.

[0010] In one embodiment, a bedding characterized by enhanced comfort, durability, long life, and decreased cost is disclosed. Bedding may include a bed sheet, bed sheeting, a pillow, a comforter, a throw, a blanket, and/or any item used to cover a mattress.

[0011] A bedding system may include sheeting in a weave pattern. The weave pattern may comprise, in some embodiments, at least one of solid and yarn-dyed woven in a weave technique selected from the group consisting of plain, dobby, Jacquard, percale, sateen, satin, twill with a 2×1 pattern, twill with a 2×2 pattern, oxford, drill, sateen with a 4-to-1 float, and satin with a 5-to-1 float. The bedding may include at least one pillowcase, a top sheet, and a fitted sheet. The sheeting may comprise at least one of a polyester warp yarn and a micro-polyester warp yarn including a denier of 15 D to 300 D and 36 filament to 388 filament. The sheeting may comprise a polyester weft yarn including multifilament separable yarns ranging with a denier of 10 D to 200 D. The sheeting may also comprise thread count of 200 TC to 3000 TC.

[0012] Embodiments may include a sheet according to any of the embodiments, and/or combinations of embodiments, and/or elements, disclosed in the present disclosure.

[0013] Some embodiments may be considered a method for a bedding system according to any of the embodiments, and/or combination of embodiments, and/or elements, disclosed in the present disclosure.

[0014] Still other embodiments may include a kit for bedding according to any of the embodiments, and/or combination of embodiments, and/or elements, disclosed in the present disclosure.

[0015] In some examples, a woven bedding fabric may include a polyester full dull separable weft yarn having a denier between 10 and 30 and a polyester warp yarn having a denier between 70 and 90. The weft yarn and the warp yarn may be woven to form a 100 percent polyester bedding fabric.

[0016] In other embodiments, a woven bedding fabric may include a semi dull separable weft yarn having a denier between 10 and 30 and a polyester warp yarn having a denier between 70 and 90. The weft yarn and the warp yarn may be woven to form a 100 percent polyester bedding fabric. The weft and/or warp yarn may be at least a semi dull yarn.

[0017] The woven bedding fabric in the weft yarn may be a full dull separable yarn including, for examples, 12 to 14 filaments. The woven bedding fabric in the warp yarn and the weft yarn may

both be a full dull yarn. The woven bedding fabric may include 10 yarns of 20 denier polyester weft separable yarn in the weft, in some examples.

[0018] The woven bedding fabric may include cooling yarns. The warp may include a cooling yarn. The weft may include a cooling yarn.

[0019] The woven bedding fabric may include a first face and a second face. The first face may be a different weave pattern than the second face. In one embodiment, the woven bedding fabric first face may be a satin woven face. The woven bedding fabric second face may be a percale woven face.

[0020] In one embodiment, the woven bedding fabric may be a contrasting face fabric, including a sateen woven face on a first face and a percale woven face on a second face. A woven bedding fabric may include a percale woven face as a top surface and a sateen woven face as a bottom surface.

[0021] The woven bedding fabric may include an application of a polymeric softener to the woven bedding fabric during construction. The woven bedding fabric may be an anti-static fabric, anti-bacterial fabric, and/or an anti-wrinkle fabric.

[0022] The woven bedding fabric may include a construction on a water jet loom.

[0023] The woven bedding fabric may include parallel inserted weft yarns.

[0024] The woven bedding fabric may include an oil percentage in the warp yarn of 2.5% or higher. The woven bedding fabric may include an oil percentage in the weft of 2.5% or higher.

[0025] The woven bedding fabric may include a construction including a loom speed of less than 600 RPM, 500 RPM, 400 RPM, 375 RPM, 350 RPM, 325 RPM, and/or 350 RPM or lower.

[0026] The woven bedding fabric may include, in certain embodiments, an oil application percentage of 2.5% or higher and a loom construction speed of less than 500 RPM.

[0027] The woven bedding fabric may include an oil application percentage between 3.0% and 3.5% and a loom construction speed of less than 400 RPM.

[0028] The woven bedding fabric may include an anti-static chemical applied at an initial dyeing stage.

[0029] Embodiments of the present disclosure may include a method for weaving a bedding fabric including inserting a polyester weft separable full dull yarn having a denier between 8 and 30; inserting a polyester warp full dull yarn having a denier between 50 and 150; forming bedding fabric including a thread count greater than 200; applying an oil percentage of at least 3 percent; and maintaining a loom construction speed of 500 RPM or less. The method may include maintaining a loom construction speed of 400 RPM or less and/or 350 RPM or less. The method may include forming a percale fabric side and a satin fabric side.

[0030] Other embodiments may be considered a system for a bedding fabric including a bedding item for covering a bed surface including a polyester full dull weft separable yarn having a denier between 10 and 30; a polyester warp full dull yarn having a denier between 70 and 90, and a thread count of at least 300. The bedding article may be formed on a water jet loom and including: an oil application percentage of at least 3 percent, and a loom construction speed of 400 RPM or less. Examples of a bedding fabric may include a thread count of at least 300, at least 400, at least 500, at least 600, at least 700, at least 800, at least 900, at least 1000 or above, and/or any thread count in between.

[0031] Examples may include, a woven bedding fabric having a cotton hand feel and appearance, but not including cotton, comprising a bedding fabric including a set of full dull warp yarns and a set of full dull weft yarns. The warp yarns may be polyester and form a first fabric surface. The weft yarns may be polyester and form an opposite fabric surface. The first fabric surface and the second fabric surface may have a different weave pattern. The weave pattern may be characterized by: a percale face formed by a first and seventh shaft percale 1/1, and a satin face formed by a second through sixth shaft satin 4/1. The woven bedding may include an application of at least a 3% oil, and an application of a loom speed of less than 400 RPM.

[0032] The inventions of the present disclosure include a method for weaving a polyester bedding fabric on a water jet loom, including: inserting an 70D to 80D polyester cooling yarn in the warp on the loom, directing a pressurized water jet to create a shed in the warp yarn, propelling a 10D to 20D separable full dull polyester weft yarn through the shed using the water jet, and applying tension to the weft yarn to secure it within the warp. The water jet may be controlled to an RPM of 400 or less and an oil percentage of at least 3.0% is applied to minimize a yarn breakage. The RPM may be 350 and include an oil percentage of at least 3.5%. The method for weaving a polyester bedding fabric may include 180 warps per inch and 72 weft insertions per inch.

[0033] A kit including a woven bedding fabric of any of the embodiments of the present disclosure is considered within the scope of this invention.

[0034] Additional aspects of the disclosed embodiment will be set forth in part in the description which follows, and in part will be obvious from the description, or may be learned by practice of the disclosed embodiments. The aspects of the disclosed embodiments will be realized and attained by means of the elements and combinations particularly pointed out in the appended claims. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the disclosed embodiments, as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0035] The accompanying drawings, which are incorporated in and constitute a part of this disclosure, illustrate various embodiments of the present disclosure. The drawings may contain representations of various trademarks and copyrights owned by the Applicant. In addition, the drawings may contain other marks owned by third parties and are being used for 5 illustrative purposes only. All rights to various trademarks and copyrights represented herein, except those belonging to their respective owners, are vested in and the property of the Applicant. The Applicant retains and reserves all rights in its trademarks and copyrights included herein, and grants permission to reproduce the material only in connection with reproduction of the granted patent and for no other purpose. 10

[0036] The drawings, photographs, pictures, and the like may contain text or captions that may explain certain embodiments of the present disclosure. This text is included for illustrative, non-limiting, explanatory purposes of certain examples and elements detailed in the present disclosure.

[0037] Embodiments of the disclosure will be better understood by a reading of the 15 Description of Embodiments along with a review of the drawings, in which:

[0038] FIG. 1 is a perspective view of a bedding set on a mattress and bed, according to an example embodiment;

[0039] FIG. 2A is a perspective view of a fitted bed sheet on a mattress and bed, according to an example embodiment;

[0040] FIG. 2B is a perspective view of a flat bed sheet on a mattress and bed, according to an example embodiment;

[0041] FIG. 2C is a diagram of the dimensions of the flat bed sheet, according to an example embodiment;

[0042] FIG. 3A is a top view of a pillowcase for a full sheet set, according to an example embodiment;

[0043] FIG. 3B is a bottom view of a pillowcase for a full sheet set, according to an example embodiment;

[0044] FIG. 3C is a diagram of the dimensions of the flat bed sheet and fitted bed sheet for a full sheet set, according to an example embodiment;

[0045] FIG. 4 is a diagram of the plain weave technique, according to an example embodiment;

[0046] FIG. 5 is a diagram of the Jacquard weave technique, according to an example embodiment;
[0047] FIG. 6 is a diagram of the percale weave technique, according to an example embodiment;
[0048] FIG. 7 is a diagram of the oxford weave technique, according to an example embodiment;
[0049] FIG. 8 is a diagram of the drill weave technique, according to an example embodiment;
[0050] FIG. 9 is a diagram of the satin weave technique, according to an example embodiment;
[0051] FIG. 10A is a table charting common sizes for items in a bedding set, according to an example embodiment;
[0052] FIG. 10B is a table charting common sizes for items in a bedding set, according to an example embodiment;
[0053] FIGS. 11A and 11B are line graphs showing one example of a weft yarn denier for various thread counts of the fabric, according to an example embodiment.
[0054] FIG. 12 shows in one example, a relationship between breakage in a fabric during weaving and a loom speed in RPM;
[0055] FIG. 13 shows in one example, a relationship between breakage in a fabric during weaving and the percent of oil applied to a warp yarn;
[0056] FIG. 14 shows in one example, a relationship between variable of oil application percentage and RPM overlaid for an acceptable breakage range in a fabric of the present disclosure; and
[0057] FIG. 15 demonstrates one example of a weaving diagram for a fabric of the present disclosure.

DESCRIPTION OF EMBODIMENTS

[0058] The following detailed description refers to the accompanying drawings. Whenever possible, the same reference numbers are used in the drawings and the following description to refer to the same or similar elements. While disclosed embodiments may be described, modifications, adaptations, and other implementations are possible. For example, substitutions, additions, or modifications may be made to the elements illustrated in the drawings, and the methods described herein may be modified by substituting reordering or adding additional stages or components to the disclosed methods and devices. Accordingly, the following detailed description does not limit the disclosed embodiments. Instead, the proper scope of the disclosed embodiments is defined by the appended claims.

[0059] It will readily be understood by one having ordinary skill in the relevant art that the present disclosure has broad utility and application. As should be understood, any embodiment may incorporate only one or a plurality of the above-disclosed aspects of the disclosure and may further incorporate only one or a plurality of the above-disclosed features. Furthermore, any embodiment discussed and identified may be considered to be part of a best mode contemplated for carrying out the embodiments of the present disclosure. Other embodiments also may be discussed for additional illustrative purposes in providing a full and enabling disclosure. Moreover, many embodiments, such as adaptations, variations, modifications, and equivalent arrangements, will be implicitly disclosed by the embodiments described herein and fall within the scope of the present disclosure.

[0060] Certain examples and elements are described herein in detail in relation to one or more embodiments, it is to be understood that this disclosure is illustrative and exemplary of the present disclosure and are made merely to provide a full and enabling disclosure. The detailed disclosure herein of one or more embodiments is not intended, nor is to be construed, to limit the scope of patent protection afforded in any claim of a patent issuing here from, which scope is to be defined by the claims and the equivalents thereof. It is not intended that the scope of patent protection be defined by reading into any claim a limitation found herein that does not explicitly appear in the claim itself.

[0061] Details with regards to examples and assemblies are provided below. Although examples and assemblies are disclosed with specific functionality, it should be understood that functionality may be shared between examples and assemblies, with some functions split between examples and

assemblies, while other functions duplicated by the examples and assemblies. Furthermore, the name of each example and element should not be construed as limiting upon the functionality of the example and element. Moreover, each component disclosed within each example and assembly can be considered independently, without the context of the other components within the same example, different examples and assemblies. Each component may contain functionality defined in other portions of this specification. In the present disclosure, each element can be claimed on its own and/or interchangeably with other elements of other examples and embodiments.

[0062] The following may depict an example of a method, of a plurality of methods, that may be performed by at least one of the aforementioned embodiments, or components thereof. Further, although the elements here may be disclosed in a particular order, it should be understood that the order is disclosed for illustrative purposes only. Elements may be combined, separated, reordered, and various intermediary stages may exist. Accordingly, it should be understood that the various assemblies, in various embodiments, may be performed/assembled in orders that differ from the ones disclosed below. Moreover, various steps may be added or removed without altering or departing from the fundamental scope of 25 the depicted methods and systems disclosed herein.

[0063] The foregoing summary, and the following detailed description provide examples and are explanatory only. Accordingly, the foregoing summary and the following detailed description should not be considered to be restrictive. Further, features or variations may be provided in addition to those set forth herein. For example, embodiments may be directed to various feature combinations and sub-combinations described in the summary and detailed description.

[0064] Embodiments of the disclosure improve upon problems of the traditional art by providing a bedding characterized by enhanced comfort, durability, long life, and decreased cost. In accordance with the present disclosure, examples of the invention provide an improved and consumer-oriented bedding system by giving the soft feel, long life and durability at a decreased cost. The bedding of the disclosure enhances user comfort, adaptability, and overall sleep quality by including the development of materials with improved thermal properties and hypoallergenic compositions that may reduce the risk of allergic reactions.

[0065] The inclusion of a wide range of weaving techniques like plain, dobby, Jacquard, percale, twill, oxford, drill, satin, and sateen allows for a greater diversity in texture, strength, and appearance. Traditional bedding often limits itself to simpler weaves, which may not offer the same level of versatility or targeted properties. The specific weave patterns are tailored to achieve desired properties such as softness, durability, and moisture-wicking, which might be less optimized in conventional bedding designs. This also improves in both the aesthetic appeal and functional aspects (like breathability, thermal regulation, etc.) of the bedding. In some examples, use of polyester and micro-polyester with specific denier and filament counts contributes to a unique balance of softness, strength, and durability. Prior bedding materials may not have optimized these properties as effectively, leading to either less durable or less comfortable options. The combination of these materials and weaving techniques could lead to more durable bedding, resulting in longer life spans and, consequently, decreased costs over time. This improvement is significant compared to traditional bedding, which might wear out more quickly or not offer the same level of comfort and durability.

[0066] Some examples of the present disclosure offer two distinct surfaces on the bedding sheeting. This can provide multifunctional use (e.g., one side for cooler feel, the other for warmth), which is a significant step up from conventional single-surface bedding. The dual surfaces, coupled with the material choice, can offer varied tactile experiences, potentially increasing comfort based on user preference or seasonal needs, an aspect not commonly addressed in prior art.

[0067] Examples of bedding of the present disclosure may enhance a sheets comfort, durability, and longevity while maintaining cost efficiency. Specialized weaving techniques and the incorporation of wrinkle resistance and antimicrobial properties may be included, offering superior tensile strength, softness, and an extended lifespan for bedding products. The bedding disclosed

herein may include high-quality polyester filament and micro polyester materials combined with various weaving techniques, which contribute to improved colorfastness standards. By specifying types of yarns and thread counts, as well as potential treatments or finishes, embodiments of the invention may ensure that bedding products retain their color better over time, even after multiple washes. This approach not only enhances the aesthetic appeal and marketability of the bedding but also addresses common consumer concerns regarding color fading and fabric degradation, thereby offering an improvement over existing textiles.

[0068] Referring now to the Figure, FIG. 1 is a perspective view of a bedding set **100** on a mattress and bed, according to an example embodiment. The bedding set includes bedding characterized by enhanced comfort, durability, long life, and decreased cost, according to an example embodiment. The bedding set may include at least one pillowcase **105**, a pillow sham **110**, a top sheet **115**, a fitted sheet **120**, a duvet cover **125** and/or various combinations thereof. The pillowcase is designed to cover and protect the pillow. It typically matches the material and design of the bedding set, ensuring aesthetic consistency. The pillowcase might be made using the same sophisticated weaving techniques as the rest of the bedding, contributing to the overall comfort and appeal. Pillowcase dimensions are crafted to snugly fit standard pillow sizes, offering ease of use and maintenance. The top sheet, also known as a flat sheet, typically lies directly underneath the blanket or duvet and on top of the sleeper. A top sheet may serve as a comfortable, hygienic layer between the sleeper and the heavier bedding above. The top sheet is often the largest piece in the set and may feature intricate weave patterns or designs, aligning with the overall theme of a bedding set. The top sheet's material and weave may contribute to the top sheet's softness, breathability, and durability, enhancing the sleeper's comfort. A fitted sheet is often designed to cover and protect the mattress. The fitted sheet may be characterized by elasticated corners, which ensure a snug and secure fit around the mattress, preventing the fitted sheet from slipping off during use. The fitted sheet is often tailored to fit various mattress sizes and depths, accommodating a wide range of bed types. A duvet cover in this invention refers to a protective and decorative casing, designed to enhance comfort, durability, and cost-effectiveness, and in some examples, forming part of a bedding set, and/or part of a bedding system. In a bedding system, a duvet cover may serve as a removable layer that encases the duvet and/or cover. A duvet cover may contribute to the overall aesthetic and tactile experience of the bedding. In some embodiments, a quilt or coverlet **130** may be included. A quilt or coverlet refers to a bedding layer that provides warmth, comfort, and aesthetic appeal, complementing the overall bedding set. The quilt or coverlet may serve as a layer for additional warmth and may also be considered a decorative element. The material and weave pattern used may contribute to the quilt's or coverlet's thermal properties, comfort level, and overall visual appeal.

[0069] With reference to FIGS. 2A-3C, the dimensions of the top sheet, the fitted sheet, and the pillowcases are discussed. FIG. 2A is a perspective view of a fitted bed sheet on a bed **200**, according to an example embodiment. FIG. 2B is a perspective view of a top sheet **115** on a bed **201**, according to an example embodiment. FIG. 2C is a diagram of the dimensions of the top sheet, according to an example embodiment. The top sheet has a first side hem length **205**, a second side hem length **210**, and a bottom side hem **215**. A hem is a finished edge on a piece of fabric or clothing. It is created by folding the edge of the fabric over and then sewing it down, thereby hiding the edges of the fabric and preventing the fabric from unraveling or fraying. In other embodiments, other types of hems, such as, but not limited to, double-fold hem and rolled hem, may be used and are within the spirit and scope of the present disclosure. The first side hem length **205** is approximately 1 inch. The second side hem length **205** is approximately 1 inch. The bottom side hem length **205** is approximately 2 inches. In other embodiments, the first side hem length **205**, the second side hem length **210**, and the bottom side hem **215** may be other lengths configured to hide the edges of the fabric. The top sheet also has a top side turn length **220** and a top portion fold length **225**. The topside turn length **220** is approximately 0.75 inches. The top portion fold length

225 is approximately 5 inches. The top sheet is folded over about the top sheet fold 230 in direction A.

[0070] FIG. 3A is a top view of a pillowcase 300 for a full sheet set, according to an example embodiment. FIG. 3B is a bottom view of a pillowcase for a full sheet set, according to an example embodiment. FIG. 3C is a diagram of the dimensions of the top sheet 301 and fitted sheet 302 for a full sheet set, according to an example embodiment. The full sheet set, as shown, is configured for a full-size bed. The pillowcase 300 includes a pillowcase length 305 a pillowcase width 310. The pillowcase length is approximately 20 inches, and the pillowcase width is approximately 30 inches. The top sheet 301 includes a top sheet length 315 and a top sheet width 320. The top sheet length is approximately 96 inches while the top sheet width is approximately 81 inches. The fitted sheet 302 includes a fitted sheet length 325 and a fitted sheet width 330. The fitted sheet length is approximately 76 inches while the fitted sheet width is approximately 54 inches. The fitted sheet may fit a mattress that is up to 16 inches thick. In other embodiments, the sheet set may include other dimensions, configured for certain bed sizes, which are within the spirit and scope of the present disclosure.

[0071] In terms of materials, exemplary sheeting may consist of either polyester or micro-polyester warp yarns, with a denier ranging from 15 D to 300 D and weft filament count between 15 D to 300. polyester, a synthetic fiber known for its strength, durability, and elasticity, may be used here as a warp yarn. Warp yarns run lengthwise, or in a longitudinal direction, in a fabric and are integral to its structure. The denier of this polyester warp yarn ranges from 15 D to 300 D. Denier is a unit of measurement that expresses fiber thickness or yarn's fineness. A range of 15 D to 300 D provides versatility in the yarn's thickness, from very fine (15 D) to relatively thick (300 D). This allows for versatility in fabric weight and texture. The filament count for this yarn, in some examples, ranges from 36 to 388. Filaments are the individual continuous strands that make up a yarn. A higher filament count typically results in a smoother, silkier fabric with enhanced strength, while a lower count leads to a coarser texture. The range from 36 to 388 allows for a broad spectrum of textures and strengths, suitable for different applications in bedding.

[0072] Micro-polyester refers to a finer version of standard polyester. It has smaller diameter fibers, which contribute to a softer, more flexible fabric. The denier range for micro-polyester warp yarn is also between 15 D to 300 D, allowing for a similar variety in fabric weight and texture as standard polyester yarn. Similarly, the filament count ranges from 36 to 388. Given the finer nature of micro-polyester, even at a higher filament count, the resulting fabric is softer and finer compared to a fabric made with a standard polyester warp yarn with a similar count.

[0073] The choice between polyester and micro-polyester warp yarns, each with its specific range of denier and filament count, allows for a broad spectrum of fabric textures and strengths. This variability ensures that the bedding is tailored to different preferences in terms of feel, durability, and appearance. The use of these specific types of yarns in the warp also significantly contributes to the overall characteristics of the bedding, such as its durability, comfort, and aesthetic appeal. In some embodiments, the warp yarn may include a combination of polyester and micro-polyester. Within the fabric weave, the polyester provides strength and durability, ensuring the fabric maintains its shape and withstands wear and tear. The micro-polyester, being finer and softer, enhances the texture of the fabric, contributing to the overall comfort and tactile appeal of the bedding. Together, they create a balance of softness and strength in the warp of the fabric. Traditional bedding fabrics might use either polyester for strength or softer materials for comfort, but the blend of both allows for a fabric that is both resilient and plush. This innovation addresses the common trade-off between durability and softness in bedding materials, offering a product that is designed to be long-lasting without sacrificing comfort.

[0074] Additionally, the sheeting may also include polyester weft yarn, which may be multifilament and separable, with a denier range of 10 D to 300 D. Polyester, a durable and versatile synthetic fiber, is chosen for its resistance to stretching, shrinking, and quick drying

properties. As weft yarns, these polyester threads run across the width of the fabric, interweaving with the warp yarns to create the fabric's structure. Multifilament yarn means that each yarn consists of many fine, continuous strands. These strands are 'separable,' meaning that they are not tightly bonded, which helps achieve certain textures or for specific post-weaving processes like texturizing.

[0075] The denier, a measure of the yarn's thickness, in some examples, may vary from 10 D to 300 D. At the lower end of this range, the yarn is finer and lighter, which results in a softer, more delicate fabric. On the higher end, the yarn is thicker and heavier, contributing to a more robust and durable fabric. This variability in denier allows for a wide range of fabric weights and textures, offering flexibility in designing bedding that caters to different preferences. The combination of these features—the durability of polyester, the intricacy of multifilament separable yarns, and the variable denier—helps define the bedding's overall feel, comfort, and performance. It enables the creation of bedding that not only meets functional requirements but also provides a pleasing tactile experience, essential for comfortable bedding.

[0076] Furthermore, the multifilament yarn may undergo the processes of intermingling and warping. Intermingling technology is a process used in textile manufacturing that enhances the cohesion and stability of filament yarns. It involves entangling the filaments at regular intervals using a jet of air, creating a more uniform and cohesive yarn structure. This technology improves the yarn's handling characteristics and reduces the need for additional processes like sizing or twisting, making it more economical and efficient for weaving or knitting. Intermingled yarns are particularly beneficial for producing high-quality fabrics with improved durability and consistency. High-Intermingle ("HIM") yarn, having 100-120 knots/meter, may be used for the warping, enhancing fabric strength and durability. The Semi-Intermingle ("SIM"), having 40-50 knots/meter, yarn may be used for warping with sizing to improve weaving efficiency and reduce yarn breakage. For the weft, SIM and/or Low Intermingle ("LIM") yarn are used to form a fabric, optimizing the balance between fabric strength and flexibility. This combination allows for the production of high-quality fabrics with specific performance characteristics. Warping involves organizing yarns in a specific order and tension onto a beam or a series of beams, preparing them to be interlaced with weft yarns to form a fabric. This process essentially may assist in ensuring that the yarns are aligned, tensioned, and ordered correctly for the weaving process, contributing to the quality and efficiency of fabric production. The polyester and/or micro-polyester sheeting may be woven in looms such as, but not limited to, projectile looms, air-jet looms, rapier looms, shuttle looms, and/or power looms.

[0077] The sheeting further may include a thread count of 200 TC to 2000 TC, which may be an aspect of the bedding's fabric quality and feel. Thread count may refer to the number of threads woven per square inch of fabric, including both warp and weft threads. At the lower end, a 200 thread count represents a fabric with a sparser distribution of threads. Bedding with this thread count is generally lighter, more breathable, and ideal for warmer climates or for those who prefer a cooler sleeping surface. It is also typically more affordable, striking a balance between quality and cost-effectiveness.

[0078] Conversely, at the high end of the spectrum, a 2000 thread count indicates a fabric with an extremely dense weave, offering a very smooth and soft texture, often regarded as luxurious. This high density of threads results in a heavier and more durable fabric, though it may be less breathable compared to lower thread count fabrics. Bedding with such a high thread count is usually more expensive, reflecting the greater amount of material used and the intricacy of the weaving process.

[0079] Spanning, for examples, from 200 to 2000 TC, this range allows for a significant variation in bedding quality, from basic to premium. As the thread count increases, the fabric typically becomes softer and more durable, but this also impacts the fabric's breathability and price. This diverse thread count range ensures that the bedding caters to various preferences and needs,

accommodating those seeking either everyday comfort or a more luxurious sleeping experience. It underscores the flexibility of the product to appeal to a wide market segment, from budget-conscious consumers to those desiring high-end, luxurious bedding.

[0080] The interaction of these elements within the bedding system contributes significantly to the overall quality of the product. The specific types of yarns, combined with the range of weave patterns, not only enhance the comfort but also may ensure the durability of the bedding. This combination may help in achieving long life and cost-effectiveness. The varied weave patterns not only add to the aesthetic appeal but also to the tactile feel of the bedding, making it versatile in design.

[0081] With reference to FIGS. 4-9, the bedding may be achieved through the use of sheeting that employs a variety of weave patterns, including plain 400, dobby, Jacquard 500, percale 600, twill 700, oxford 800, drill, and satin 900. Each weave pattern may include warp yarn 405 and weft yarn 410. In some embodiments, other weave patterns may be used and are within the spirit and scope of the present disclosure. The weave pattern of the bedding encompasses a diverse and sophisticated array of weaving techniques, combining both solid and yarn-dyed fabrics. Solid fabrics refer to materials made from yarns of a single color, resulting in a uniform and consistent appearance. In contrast, yarn-dyed fabrics involve coloring the yarns prior to weaving, which allows for a multitude of colors and intricate patterns in the final fabric.

[0082] The plain weave is the simplest form, characterized by warp and weft yarns crossing each other at right angles, creating a basic crisscross pattern. In this weave, each weft thread crosses the warp threads by going over one, then under the next, with each row alternating to create a uniform and tight structure. This is contrasted by the dobby weave, which uses a special loom to produce small, geometric designs, offering more textural diversity. The Jacquard weave, on the other hand, is a complex method that employs a Jacquard loom, facilitating the creation of elaborate patterns like brocades and damasks. The percale weave, a type of plain weave, is notable for its high thread count, yielding a fabric that is crisp, smooth, and durable, often favored in premium bed linens.

[0083] Twill weave is easily recognized by its distinctive diagonal ribs or lines, providing strength and a good drape to the fabric. In some embodiments, the weave technique may include twill with a 2×1 pattern. In a 2×1 twill pattern, the weave follows a sequence where the weft thread goes over two warp threads and then under one warp thread. This pattern creates a diagonal ribbing on the fabric's surface. In bedding, the 2×1 twill weave may contribute to both the aesthetic and functional aspects of the fabric. The diagonal pattern is not only visually appealing but also may enhance the durability and drape of the material. It provides a balance of softness and strength, making it suitable for comfortable yet long-lasting bedding. The 2×1 twill weave offers a denser and more robust fabric with a distinct texture. It typically shows less dirt and wear, making it more suitable for items that require frequent use and washing, like bedding. The diagonal lines also may add an element of design and sophistication to the fabric's appearance, which is an aesthetic improvement over simpler weave patterns. In some embodiments, the weave technique may include twill with a 2×1 pattern. In a 2×2 twill pattern, the weave sequence involves the weft thread passing over two warp threads and then under two warp threads. This pattern creates a more pronounced diagonal rib than a 2×1 twill, leading to a distinct texture. The 2×2 twill pattern provides a balance of flexibility and strength, making the fabric more durable and suitable for bedding that undergoes regular use and washing. The diagonal ribbing also adds a subtle aesthetic appeal to the bedding. The 2×2 twill pattern offers a stronger and more durable fabric compared to simpler weaves like the plain weave. The enhanced diagonal ribbing provides a more textured and visually appealing surface. This weave structure also tends to be more resistant to wrinkles and maintains its shape better, which is particularly advantageous for bedding applications.

[0084] Oxford weave offers a basketweave structure, a variation of the plain weave, known for its slightly rough, textured feel, commonly seen in dress shirts. In an Oxford weave, multiple weft threads are crossed over an equal number of warp threads, creating a symmetrical, checkerboard-

like pattern. This method results in a fabric that is relatively thick and soft, with a slightly textured surface that is both aesthetically pleasing and functional. Furthermore, the drill weave, a form of twill, produces a heavier fabric with diagonal ribbing, typically used in work clothing due to its durability. This weaving technique involves passing the weft thread over one or more warp threads and then under two or more warp threads, with a “step” or offset between rows to create the characteristic diagonal lines. Drill fabric is known for its strength, durability, and resistance to tearing, making it suitable for heavy-duty use.

[0085] Lastly, the satin weave is distinguished by its smooth, glossy surface, achieved by floating warp yarns over weft yarns, lending a luxurious sheen to the fabric. In some embodiments, the weave technique may include satin with a 4-to-1 float. In satin with a 4-to-1 float, the weave pattern involves one weft thread floating over four warp threads and then going under one warp thread. This pattern creates long floats on the fabric surface, contributing to its high luster and smooth texture. In bedding, the 4-to-1 float in satin provides a luxurious feel and appearance. The longer floats create a reflective surface, giving the fabric a glossy look. This weave pattern also contributes to the fabric's drape, making it suitable for more elegant and high-end applications. Compared to other weaves, the 4-to-1 float in satin offers a more luxurious and refined appearance due to its high sheen. This makes it desirable for high-quality bedding, evening wear, and other applications where a premium look and feel are essential. The smooth surface also means less friction, which is beneficial for bedding as it is gentler on skin and hair. This weave is an improvement in terms of aesthetic appeal and tactile experience over less lustrous fabrics.

[0086] In some embodiments, the weave technique may include satin with a 5-to-1 float. In a 5-to-1 float satin, the pattern involves one weft thread floating over five warp threads before going under one warp thread. This creates even longer floats on the fabric surface compared to a 4-to-1 float, enhancing the luster and smoothness of the fabric. The 5-to-1 float in satin is particularly significant in bedding, where a highly lustrous and elegant appearance is desired. The extended floats produce a very reflective surface, contributing to the fabric's rich sheen and smooth hand feel. This weave also affects the drape of the fabric, making it fluid and graceful. The longer floats result in a higher sheen and a more luxurious feel compared to fabrics with shorter floats, such as 4-to-1 satin. This makes it ideal for premium applications where the visual and tactile qualities of the fabric are paramount. The smooth surface is also beneficial for reducing friction, which is especially valuable in high-end bedding and apparel, offering enhanced comfort and a gentler interaction with skin and hair.

[0087] Overall, the combination of these weaving techniques in the bedding design allows for an impressive blend of functionality, comfort, and aesthetic appeal. The use of both solid and yarn-dyed fabrics, alongside the variety in weaving styles, ensures that the bedding meets a wide range of consumer preferences, offering everything from durability and smoothness to intricate designs and luxurious textures.

[0088] Referring now to FIGS. 10A and 10B, a table charting common sizes for items in a bedding set is shown, according to an example embodiment. Chart 1000 lists the sizes for various sheet set components for different bed sizes, including Twin, Full, Queen, King, and California King. The components listed are flat sheets, fitted sheets, and pillowcases, with each item's finish size provided in inches. For instance, the finish size of a Twin flat sheet is specified as “66×96+4””, indicating its width and length, plus an additional length (for tucking and/or decorative purposes). Similarly, the fitted sheet sizes and pillowcase sizes are provided, with the fitted sheets having a “+15” designation, likely indicating the depth of the mattress they can cover, and the pillowcases having a “+4” designation, for an extra flap or closure feature. These sizes are common but can vary according to specific requirements.

[0089] Chart 1001 includes exemplary finish sizes for duvet cover and comforter set sizes. This section provides finish sizes in inches for comforter and duvet cover sizes across various bed types, including Twin, although the full range of bed sizes covered in this section isn't fully visible in the

provided output. As with the sheet sets, the sizes are presented as common dimensions, subject to variation based on particular needs or designs. These sizes are mostly common but can vary, implying customization or different standards might apply. Both charts emphasize the variability and customization available in bedding sizes, catering to a wide range of bed dimensions and customer preferences. The detailed size specifications help consumers select bedding that fits their specific needs. However, sets with different dimensions may be used and are within the spirit and scope of the present disclosure.

[0090] With reference to FIGS. **11A** and **11B**, a line graph is shown demonstrating examples of a yarn denier for various thread counts of the fabric, according to an example embodiment. In one example, a lower thread count fabric may include higher denier filament yarns and for a higher thread count fabric, lower denier may be include.

[0091] FIG. **12** shows in one example, a relationship between breakage of the fabric during weaving and a loom speed in RPM. As loom speed RPM increases, as does breakage.

[0092] FIG. **13** shows in one example, a relationship between breakage in a fabric during weaving and the percent of oil applied to a warp yarn. In an inverse relationship, as oil percent increases, breakage decreases.

[0093] Also, referencing FIG. **14**, in an inverse relationship, yarn with a lower RPM loom speed includes less breakage, and yarn including an increased oil application includes less breakage. The overlap between the two variable forms, in one embodiment, an acceptable range of breakage, minimizing the amount of deviation from traditional yarn applications by specific applications of both variables.

[0094] FIG. **15** demonstrates one example of a weaving diagram for a fabric of the present disclosure.

[0095] In examples, woven fabric has two sidewise yarns used in the fabric's construction. One side of woven fabric is the warp yarn, and the other is called weft yarn. Both warp and weft yarn are interconnected in the formation of cloths and/or fabric, by way of example fabric used for bedding, such as by example, top sheets, fitted sheets, pillow cases, and/or duvets and duvet covers.

[0096] In certain examples, inventions of the present disclosure may include a fabric including a high thread count, good feel to hand, and/or economic pricing, without incurring raised costs to produce, for example such as cotton or CVC sheets, however includes a more substantial look and feel than lower cost products, for example like, polyester Micro Fiber. The fabric may include warp yarn. The warp yarn may, by way of example, be a denier filament yarn and/or a polyester spun yarn. A weft yarn, in one example, may be a separable denier filament yarn with multi-ply yarns. The fabric may be a 100% polyester construction. The fabric may retain a thread count which is testable in a lab. The 100% polyester construction may be able to be made in most any weave, such as for example, those various weaves explained herein. The fabric may be brushed and/or non-brushed. The fabric may include a cooling yarn in the warp. In some examples, a warp yarn range may be from 50 denier to 150 denier. The fabric may include a separable full dull denier filament, multi-ply yarn in the weft. In some examples, the weft yarn may be from 8 denier to 30 denier and/or with parallel yarn. In one embodiment, the weave may be a sateen weave. The fabric may include a first face and a second face. The first face may be a sateen woven face. The second face may be a percale woven face.

[0097] In some instance, a polymeric softener may be included to produce and impart in the fabric an anti-static feature, anti-bacterial feature, and/or an anti-wrinkle feature. An application of polymeric softener may reduce friction between yarns in a fabric and also increase the hand feel. Examples of polymeric softeners may include silicone-based polymers, acrylic polymers, and/or silicone based polymers. In one example, a BGX chemical, polymeric softener may be included.

[0098] Embodiments for a bedding item may include a cooling yarn in the warp, (by way of one example, a Kooltex yarn by Reliance Industries). The bedding item (fabric) may include a separable yarn in the weft. The separable yarn may be a 100% separable yarn. The cooling yarn

may be between 70 denier and 90 denier, and in some examples 80 denier. The separable yarn may be between 10 denier and 30 denier, and in some examples 20 denier. A weft insertion may include 10 yarns of 20 denier separable polyester yarn. The bedding item may be, in one embodiment, an 80D (denier)×20D construction. There may be 180 warps per inch and 72 weft insertions per inch in certain embodiments (180×72×10=90).

[0099] Combining certain yarns in the weft and warp together may present challenges. Typically, 100% polyester fabric is made on water jet looms for cost effective measures. Water jet looms are typically high-speed weaving machines with high production speed for producing large volumes of fabric quickly, maximizing production output. A typical water jet loom may operate at 700 to 1200 RPM. Applicant realized that a drawback of water jet looms is that they do not have an accumulator, such that pick insertion is not available, thus parallel yarns are selected. To applicant's knowledge separable denier filament yarn traditionally is not used in the weft on water jet looms due to these challenges.

[0100] Additionally, to applicant's knowledge, in separable denier filament yarn spinners, a full dull yarn has not prior been made. Machinery and method changes may be made to accommodate the construction of separable denier filament yarn on water jet looms and also to accommodate full dull yarn production. With parallel yarns, applicant realized that use of more filaments in one yarn created challenges with snarl in the fabric. Additionally, challenges occurred with breakages in the fabric during the weaving process, failing to produce the desired fabric softness. Still other challenges arose with use of a full dull yarn in finer counts due to the dull yarn not coming onto the bobbins. Applicant overcame these and other challenges by increasing an oil percentage in the warp yarn and/or, in some examples, introducing a decrease of the speed of the looms to address and ensure minimal breakages in the fabric.

[0101] In one embodiment, a fabric may include an increased oil percentage in the warp yarn of higher than 2.0%. The warp yarn may include an oil percentage greater than 2.5%. In other examples, a fabric may include an increase of at least one oil percentage. In still other examples a warp yarn may include an applied oil percentage of at least 3%, at least 3.5%, and/or greater than 3.25%. In some examples, an oil percentage of between 2.5% and 3.5% may be included. An oil may be considered, by way of example, a coning oil.

[0102] Examples in other embodiments may include a loom speed of less than 600 RPM, a loom speed reduction of at least 100 RPM, a loom speed of 550 or less, and/or a loom speed of 500 or less, and or a loom speed of 450 or less, a loom speed of 400 or less, a loom speed of 350 or less, and/or any combinations thereof.

[0103] In some examples, a fabric may include an oil percentage of at least 2.5% to 3.5% and weaving loom speed controlled at no more than 550 RPM, 500 RPM, 400 RPM, or 350 RPM.

[0104] Static in the fabric was also recognized as a challenge. In some examples, such challenge was overcome by including, in an initial dyeing, a polymeric softener.

[0105] Embodiments may include a fabric having a polyester yarn that is full dull. Including a full dull yarn may produce a look more similar to a cotton fabric. One example of a bedding fabric may include a polyester full dull yarn. The full dull yarn may be in the weft. The full dull yarn may be in the warp. In one embodiment, a bedding fabric may include a polyester full dull, separable, and/or about 20 denier yarn (with acceptable deniers ranging between 15 to 25 denier). Full dull yarns are created by adding chemicals to the polyester when it is formed (often referenced as a matting agent). In some examples, a dulling agent or delustrant may be applied to the yarn to form a full dull yarn. In some examples, between 2.0% and 3.0% of a matting agent may be applied. In other examples, 2.0% to 2.9% of a matting agent may be applied. Full dull yarns may be included in a bedding fabric.

[0106] In one example, an oil percentage may be increased with moisture wicking yarn. In some examples, polymerizing PTA and MEG raw material for a chemical reaction at 90 to 300 degrees centigrade may be beneficial. A delustrant substance may be included to make the yarn full dull.

[0107] In the weaving process, a RPM may be reduced from about 650 to 350 RPM in forming a woven fabric.

[0108] In a cooling yarn component of a fabric, moisture may be spread over a larger surface area on the fabric enabling moisture to evaporate more quickly and dry fast.

[0109] Embodiments may include a bedding fabric including a cooling yarn in the warp and full dull yarn in the weft, the bedding fabric including a cotton like feel. A dulling agent may be included with the fabric construction to impart a dull feel and look similar to cotton. The fabric may include a high thread count, a permanent cooling, an anti-static surface, be wrinkle free, be hypoallergenic, be anti-microbial, more highly durable, and/or include a stay fit. In some examples, the inventions of the present disclosure may be considered a 100% polyester sheet. In some instances, the 100% polyester sheet may include at least a 200 thread count or higher. In some instance, a thread count of 300 or higher may be considered a high thread count. A 100% polyester, cooling warp yarn may be combined with a 100% polyester weft separable yarn in some examples of a bedding fabric.

[0110] In some examples, a bedding fabric may include an 80D Kooltex warp, a 20D separable, full dull poly weft, the fabric including $180 \times (72 \times 10) = 900$). In other examples, a warp yarn may be a cooling yarn from about 50D to about 150D. The weft yarn may be a separable full dull yarn having about 15D and/or above. The yarn may include 12 to 14 filaments. The warp yarn may form a top face of the fabric. The fabric may include a RPM of not more than 400 to prevent breakage. A polymeric softener may be applied to impart anti-static, anti-bacterial, soft hand feel, and/or anti-wrinkle features. An oil percentage of 2.5% to 3.5% may be applied for dull yarn and lack of slip in the reed.

[0111] Embodiments may include a bedding sheet including a satin surface and a percale surface.

[0112] In some examples, a bedding fabric may include a cooling warp yarn from 50D (denier) to 15D and a separable full dull weft yarn from 15D and above. In some examples, the yarn may include from 12 to 14 filaments. The cooling warp yarn may form an upper side of the bedding fabric. The bedding fabric may include a RPM of not exceeding 450. A polymeric softener may be included. An oil percentage may be included of between 2.5% and 3.5%. An oil percentage application of greater than 3.5% may create challenges of slipping in the reed.

[0113] Referencing FIG. 15, #1 shaft and #7 shaft may form a percale 1/1 bottom side. #2-#6 shaft and #8-#12 shaft may form a satin 4/1 upper side. The bedding fabric may include a satin upper side and a percale bottom side. The warp may form a sateen side of and the weft forms the other side.

[0114] Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.

[0115] In some examples, a warp yarn may be a denier filament yarn or a polyester spun yarn. A weft yarn may be a separable denier filament yarn with multi ply yarns. A bedding fabric may be brushed and/or non-brushed. In some embodiments, a bedding fabric may include at least a 500 TC (thread count), a 600 TC, a 700 TC, an 800 TC, and/or at least a 900 TC.

[0116] While water jet looms are specifically mentioned, it is contemplated that any type of loom may be used in construction of a bedding fabric according to embodiments of the invention. Bedding fabrics constructed on air jet and other types of looms are within the scope of this disclosure. Bedding fabrics, in some examples, may include semi dull yarns on a top face, a bottom face, and/or both faces. One bedding fabric face may include a semi dull yarn, while the other face includes a full dull yarn. In other embodiments, one face may be semi dull or full dull, while the other face is semi dull, full dull or neither.

[0117] Numerous characteristics and advantages have been set forth in the foregoing description, together with details of structure and function. Many of the novel features are pointed out in the

appended claims. The disclosure, however, is illustrative only, and changes may be made in detail, especially in matters of shape, size, and arrangement of parts, within the principle of the disclosure, to the full extent indicated by the broad general meaning of the terms in which the general claims are expressed. It is further noted that, as used in this application, the singular forms “a,” “an,” and “the” include plural referents unless expressly and unequivocally limited to one referent.

[0118] Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the invention are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard deviation found in their respective testing measurements. Moreover, all ranges disclosed herein are to be understood to encompass any and all subranges subsumed therein, and every number between the end points. For example, a stated range of “1 to 10” should be considered to include any and all subranges between (and inclusive of) the minimum value of 1 and the maximum value of 10; 25 that is, all subranges beginning with a minimum value of 1 or more, e.g. 1 to 6.1, and ending with a maximum value of 10 or less, e.g., 5.5 to 10, as well as all ranges beginning and ending within the end points, e.g. 2 to 9, 3 to 8, 3 to 9, 4 to 7, and finally to each number 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10 contained within the range.

[0119] Any sequence(s) and/or temporal order of steps of various processes or methods that are described herein are illustrative and not restrictive. Accordingly, it should be understood that, although steps of various processes or methods may be shown and described as being in a sequence or temporal order, the steps of any such processes or methods are not limited to being carried out in any particular sequence or order, absent an indication otherwise. Indeed, the steps in such processes or methods generally may be carried out in various different sequences and orders while still falling within the scope of the present invention. Accordingly, it is intended that the scope of patent protection is to be defined by the issued claim(s) rather than the description set forth herein.

[0120] Elements and terms used herein refer to that which one of ordinary skill would understand such a term to mean based on the contextual use of the term herein. To the extent that the meaning of a term used herein—as understood by the ordinary artisan based on the contextual use of such term—differs in any way from any particular dictionary definition of such term, it is intended that the meaning of the term as understood by one of ordinary skill should prevail.

[0121] Regarding applicability of 35 U.S.C. § 112, ¶6, no claim element is intended to be read in accordance with this statutory provision unless the explicit phrase “means for” or “step for” is actually used in such claim element, whereupon this statutory provision is intended to apply in the interpretation of such claim element.

[0122] All rights, including copyrights herein, are vested in and the property of the Applicant.

[0123] The specification includes examples, the disclosure's scope is indicated by the following claims, and furthermore, while the specification has been described in language specific to structural features and/or methodological acts, the claims are not limited to the features or acts described above. Rather, the specific features and acts described above are disclosed as examples for embodiments of the disclosure. Insofar as the summary and description above and the accompanying drawing disclose any additional subject matter that is not within the scope of the claims below, the disclosures are not dedicated to the public and the right to file one or more applications to claims such additional disclosures is reserved.

Claims

1. A woven bedding fabric comprising: a polyester at least semi dull separable weft yarn having a denier between 10 and 30, a polyester warp yarn having a denier between 70 and 90, wherein the weft yarn and the warp yarn are woven to form a 100 percent polyester bedding fabric.
2. The woven bedding fabric of claim 1 including a weave pattern, the weave pattern comprising at least one of solid and yarn-dyed woven in a weave technique selected from a group consisting of

plain, dobby, Jacquard, percale, twill with a 2×1 pattern, twill with a 2×2 pattern, oxford, drill, satin, sateen, satin with a 4-to-1 float, and satin with a 5-to-1 float.

3. The woven bedding fabric of claim 1 wherein the weft yarn is a full dull separable yarn including 12 to 14 filaments.

4. The woven bedding fabric of claim 3 wherein the warp yarn and the weft yarn are both a full dull yarn.

5. The woven bedding fabric of claim 4 wherein the warp yarn is a cooling yarn.

6. The woven bedding fabric of claim 5 including a first face and a second face, wherein the first face is a different weave pattern than the second face.

7. The woven bedding fabric of claim 6 wherein the first face is a satin woven face.

8. The woven bedding fabric of claim 7 wherein the second face is a percale woven face.

9. The woven bedding fabric of claim 1 wherein the percale woven face is a top surface and the sateen woven face is a bottom surface.

10. The woven bedding fabric of claim 1 including an application of a polymeric softener to the woven bedding fabric during construction.

11. The woven bedding fabric of claim 10 wherein the bedding fabric is an anti-static fabric, anti-bacterial fabric, and/or an anti-wrinkle fabric.

12. The woven bedding fabric of claim 1 wherein the polyester weft separable yarn includes 12 to 14 filaments.

13. The woven bedding fabric of claim 1 wherein the bedding fabric includes 10 yarns of 20 denier polyester weft separable yarn in the weft.

14. The woven bedding fabric of claim 12 wherein the fabric is constructed on a water jet loom.

15. The woven bedding fabric of claim 14 wherein the weft yarns are parallel inserted weft yarns.

16. The woven bedding fabric of claim 1 including an oil percentage in the warp yarn of 2.5% or higher.

17. The woven bedding fabric of claim 1 constructed with a loom speed of less than 600 RPM.

18. The woven bedding fabric of claim 1 including an oil application percentage of 2.5% or higher and a loom construction speed of less than 600 RPM.

19. The woven bedding fabric of claim 1 including an oil application percentage between 3.0% and 3.5% and a loom construction speed of less than 550 RPM.

20. The woven bedding fabric of claim 19 including an anti-static chemical applied at an initial dyeing stage.

21. The woven bedding fabric of claim 1 wherein bedding includes a pillowcase, a pillow wham, a top sheet, a fitted sheet, a cover, a duvet, a duvet cover, a blanket, and any combination thereof.

22. A method for weaving a bedding fabric comprising: inserting a polyester weft separable full dull yarn having a denier between 8 and 30, inserting a polyester warp full dull yarn having a denier between 50 and 150, forming bedding fabric including a thread count greater than 200, applying an oil percentage of at least 3 percent, maintaining a loom construction speed of 500 RPM or less.

23. The method of claim 22 including forming a percale fabric side and a satin fabric side.

24. A system for a bedding fabric comprising: a bedding item for covering a bed surface including: a polyester at least semi dull weft separable yarn having a denier between 10 and 30, a polyester warp at least semi dull yarn having a denier between 70 and 90, a thread count of at least 300, the bedding article formed on a water jet loom and including: an oil application percentage of at least 3 percent, and a loom construction speed of 550 RPM or less.

25. A woven bedding fabric having cotton characteristics, but not including cotton, comprising: the fabric including a set of at least semi dull warp yarns and a set of at least semi dull weft yarns, said warp yarns being polyester and forming a first fabric surface, said weft yarns being polyester and forming an opposite fabric surface, wherein said first fabric surface and said second fabric surface are different weave patterns, said weave pattern characterized by: a percale face formed by a first

and seventh shaft percale 1/1, and a satin face formed by a second through sixth shaft satin 4/1.

26. The woven bedding fabric of claim 25 including: an application of at least a 3% oil, and an application of a loom speed of less than 550 RPM.

27. A method for weaving a polyester bedding fabric on a water jet loom, comprising: inserting an 70D to 80D polyester cooling yarn in the warp on the loom, directing a pressurized water jet to create a shed in the warp yarn, propelling a 10D to 20D separable full dull polyester weft yarn through the shed using the water jet, applying tension to the weft yarn to secure it within the warp, wherein the water jet is controlled to an RPM of 550 or less and an oil percentage of at least 3.5% is applied to minimize a yarn breakage.
